

NOVYY USHTAGAN, Kazakhstan

WMO: 346910

Lat: 47.9056N

Lon: 48.7906E

Elev: -10

StdP: 101.45

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.0	-19.9	-26.4	0.3	-22.6	-23.3	0.5	-19.3	10.0	-6.3	8.9	-5.3	2.5	320	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	38.1	19.9	36.4	19.6	34.6	18.9	21.5	33.7	20.7	32.6	19.9	31.7	3.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	12.8	24.5	16.7	11.9	24.5	15.6	11.1	24.7	62.4	33.4	59.4	32.6	56.9	31.8	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	9.9	8.3	7.6	DB	-25.1	40.3	4.9	1.4	-28.6	41.3	-31.4	42.1	-34.2	42.9	-37.7	43.9
(5)				WB	-25.4	23.1	4.7	1.4	-28.8	24.1	-31.5	24.8	-34.1	25.6	-37.5	26.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	-6.6	-5.6	2.1	11.3	18.9	24.0	26.5	25.0	17.7	9.6	1.1	-4.2	(6)
(7)		DBStd	12.91	6.27	6.75	4.67	4.80	4.75	4.02	3.31	4.16	4.47	4.82	5.78	5.77	(7)
(8)		HDD10.0	2021	515	437	248	41	2	0	0	2	67	269	439	(8)	
(9)		HDD18.3	3776	773	671	504	216	52	5	0	3	63	273	518	698	(9)
(10)		CDD10.0	2044	0	0	2	79	277	421	512	465	233	54	2	0	(10)
(11)		CDD18.3	758	0	0	0	4	68	176	254	210	44	2	0	0	(11)
(12)		CDH23.3	9548	0	0	0	72	822	2148	3242	2718	526	19	0	0	(12)
(13)		CDH26.7	4840	0	0	0	13	339	1080	1753	1471	182	2	0	0	(13)
(14)	Wind	WSAvg	3.0	3.1	3.4	3.5	3.3	3.2	2.9	2.6	2.7	2.8	2.9	3.1	(14)	
(15)	Precipitation	PrecAvg	199	16	12	14	17	23	23	13	11	16	20	16	17	(15)
(16)		PrecMax	300	45	50	50	39	70	58	52	35	45	86	37	39	(16)
(17)		PrecMin	139	1	0	0	0	2	1	0	0	0	1	1	1	(17)
(18)		PrecStd	42	10	11	13	12	16	16	14	9	13	18	9	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	8.2	17.1	28.6	35.0	39.0	40.4	39.7	33.9	25.6	15.3	7.9	(19)
(20)			MCWB	4.8	5.4	9.4	15.4	17.2	19.3	20.3	20.5	17.9	15.2	10.7	6.0	(20)
(21)		2%	DB	4.0	5.2	14.1	25.4	32.6	36.7	38.1	37.8	31.0	22.7	12.5	6.0	(21)
(22)			MCWB	2.8	3.0	7.9	14.2	17.1	19.2	20.0	20.4	17.1	13.6	9.6	4.9	(22)
(23)		5%	DB	2.2	3.5	11.3	22.7	30.6	34.7	36.4	36.0	28.7	19.9	10.4	4.1	(23)
(24)			MCWB	1.1	1.7	6.4	13.0	16.4	18.7	19.7	19.7	15.8	12.3	8.1	2.8	(24)
(25)		10%	DB	0.8	2.1	8.9	20.1	28.1	32.6	34.4	34.0	26.3	17.4	8.4	2.5	(25)
(26)			MCWB	-0.2	0.7	5.0	11.7	15.7	17.9	18.9	18.9	15.0	11.6	6.6	1.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.8	5.7	10.3	16.4	19.6	22.0	22.6	22.9	19.7	16.7	12.1	6.6	(27)
(28)			MCDWB	6.1	7.8	15.1	25.4	30.9	33.9	35.3	35.4	29.7	23.1	13.8	7.3	(28)
(29)		2%	WB	2.9	3.4	8.3	14.9	18.3	20.8	21.5	21.5	18.2	14.7	10.1	4.7	(29)
(30)			MCDWB	3.8	4.8	13.2	23.8	28.5	32.0	33.7	33.9	27.4	20.0	12.0	5.8	(30)
(31)		5%	WB	1.2	1.9	6.8	13.7	17.3	19.7	20.7	20.6	17.0	13.3	8.3	3.0	(31)
(32)			MCDWB	2.0	3.1	10.8	21.3	27.4	30.9	32.6	32.9	25.9	18.6	10.0	4.0	(32)
(33)		10%	WB	0.0	0.8	5.3	12.4	16.4	18.9	20.0	19.7	16.0	11.8	6.6	1.5	(33)
(34)			MCDWB	0.8	1.9	8.4	19.1	26.3	29.7	31.6	31.9	24.3	16.9	8.4	2.4	(34)
(35)	Mean Daily Temperature Range	MDBR	6.5	7.8	8.9	11.7	12.1	12.3	12.3	13.4	12.7	10.8	7.4	5.9	(35)	
(36)		5% DB	MCDBR	5.3	7.6	12.6	15.3	15.0	14.9	14.7	15.7	16.0	14.7	9.8	5.5	(36)
(37)			MCWBR	4.7	5.6	7.4	7.1	5.7	5.4	5.0	5.6	6.6	7.4	6.6	4.6	(37)
(38)		5% WB	MCDBR	5.0	6.3	11.6	14.1	13.2	12.9	13.0	13.9	13.2	12.7	8.5	5.3	(38)
(39)			MCWBR	4.6	5.0	7.3	7.1	5.9	5.3	5.2	5.7	6.4	7.5	6.5	4.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.350	0.373	0.410	0.474	0.453	0.425	0.431	0.441	0.403	0.375	0.354	0.351	(40)	
(41)		taud	2.093	2.048	2.081	1.961	2.094	2.244	2.238	2.182	2.238	2.257	2.217	2.109	(41)	
(42)		Ebn at Noon	672	750	790	776	813	840	829	800	794	750	673	614	(42)	
(43)		Edh at Noon	92	120	136	169	154	134	133	134	116	96	81	81	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.08	2.07	3.32	4.80	6.14	6.71	6.55	5.76	4.31	2.61	1.30	0.77	(44)	
(45)		RadStd	0.24	0.35	0.33	0.35	0.46	0.54	0.40	0.34	0.31	0.27	0.19	0.14	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (3 neighbors)	N/A	N/A	N/A	+1.52	N/A	-0.73	N/A	N/A	+170	+115	

Nomenclature: See separate page